

Denominator-Adjusted Analysis of COVID-19 VAERS Reporting Patterns

Using Public VAERS and CDC Lot Distribution Data

**Prepared by:
Independent Researcher
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This report presents a denominator-adjusted analysis of publicly available VAERS and CDC lot-distribution data. The analysis identifies reporting patterns for further investigation and does not establish causation, clinical incidence, or product defects.

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Executive Summary

This report examines COVID-19 vaccine adverse-event reports submitted to the Vaccine Adverse Event Reporting System (VAERS) and combines those records with publicly released CDC lot-distribution data to evaluate reporting patterns at the manufacturer and vaccine-lot level.

This analysis examined whether apparent lot-level reporting patterns remained visible after accounting for distribution volume.

Raw VAERS report counts alone provide an incomplete basis for lot-level comparison. A vaccine lot associated with a large number of reports may simply have been distributed more broadly than other lots. Without denominator information (some measure of how many doses were distributed or administered), reliable lot-level comparisons are limited.

This project incorporated publicly released CDC lot-distribution records, allowing calculation of denominator-adjusted reporting metrics for matched Pfizer lots. These included reports per million doses shipped, serious reports per million doses shipped, death reports per million doses shipped, expected serious reports, observed-versus-expected serious-reporting ratios, and approximate confidence intervals.

Multiple apparent lot-level signals diminished after denominator adjustment, indicating that distribution volume explained some raw-count differences.

Certain lots remained elevated after denominator adjustment. The largest denominator-adjusted serious-reporting elevation observed in this analysis was Pfizer lot GJ3277, which showed an observed-versus-expected serious-reporting ratio of 5.049 with an approximate 95% confidence interval of 4.638 to 5.460. The EN6200, EN6201, and EN6202 lots also remained elevated, with observed-versus-expected ratios of 2.719, 2.855, and 2.693, respectively.

Within the matched Pfizer lots analyzed, adjustment for shipped-dose volume reduced some apparent signals while others remained elevated.

The observed differences do not establish causation, clinical incidence, manufacturing defects, or product-quality differences. Nor do they exclude those possibilities. They indicate that shipped-dose adjustment materially altered the interpretation of lot-level reporting differences and identified certain lots as appropriate candidates for additional investigation using stronger independent datasets.

Introduction

Since the introduction of the COVID-19 vaccines, questions have been raised regarding the interpretation of adverse-event reports submitted to the Vaccine Adverse Event Reporting System (VAERS). Critics of many early analyses argued that raw report counts alone were insufficient because they lacked denominator information. A vaccine lot associated with a large number of reports may simply have been distributed more widely than other lots.

The criticism reflected a legitimate methodological challenge. Without information regarding distribution volume, reliable comparison between vaccine lots is limited. Higher report counts do not necessarily indicate higher reporting rates.

The release of CDC lot-distribution data provided an opportunity to examine the issue more directly. By combining VAERS reports with publicly available lot-distribution records, it became possible to evaluate whether apparent reporting differences remained visible after adjustment for shipped-dose volume.

This report presents the results of that denominator-adjusted analysis. The objective is not to determine causation, estimate clinical incidence, or evaluate individual adverse-event reports. Rather, the analysis seeks to determine whether adjustment for distribution volume materially changes the interpretation of apparent lot-level differences.

Researcher Disclosure

I approached this project with questions regarding the transparency of COVID-era public-health decision making and vaccine-safety monitoring. That perspective contributed to my interest in the subject and my decision to explore the available data.

The purpose of this analysis was not to confirm a predetermined conclusion. The objective was to evaluate whether **apparent reporting differences** persisted after denominator information became available through publicly released CDC lot-distribution records.

The findings presented in this report reflect the results of the analysis performed. Readers should evaluate the methodology, assumptions, calculations, limitations, and conclusions on their own merits.

Objectives

The objectives of this project were:

1. Evaluate manufacturer-level reporting patterns within COVID-19 VAERS data.
2. Evaluate lot-level reporting patterns.
3. Incorporate publicly released CDC lot-distribution data where available.
4. Calculate denominator-adjusted reporting metrics.
5. Calculate observed-versus-expected serious-reporting ratios.
6. Estimate approximate confidence intervals for elevated lot-level findings.
7. Produce a transparent and reproducible analytical framework using publicly available data.

Data Sources

The analysis used the following data sources:

- Public VAERS report data
- Public VAERS vaccine data
- Public VAERS symptom data
- Publicly released CDC lot-distribution records
- Manufacturer fields
- Vaccine lot identifiers
- Serious adverse-event indicators
- Death indicators
- Hospitalization, life-threatening, and disability indicators
- Symptom records

Data processing and analysis were performed using DuckDB, Python, Pandas, and related statistical libraries. The repository, scripts, output files, and documentation were publicly released to support independent review.

Data Availability and Access Challenges

One of the practical obstacles encountered during this project involved access to lot-level distribution information.

During the course of the research, I attempted to obtain vaccine lot information through the CDC's COVID-19 Vaccine Lot Number and Expiration Date system. **Access to that public system was denied.**

More important than the denial itself was its practical consequence. The absence of lot-level distribution data was one of the principal constraints affecting earlier lot-level analyses. Reliable comparison of vaccine lots requires both a numerator (reported events) and a denominator (distribution volume or doses administered).

For several years, this limitation represented one of the strongest criticisms of lot-based VAERS investigations. In many cases, that criticism was justified.

The denominator-adjusted portion of this project became possible only after publicly released lot-distribution records became available. Access to denominator information remains an important practical consideration in the ability of independent researchers to evaluate competing interpretations of lot-level reporting differences.

Data and Code Availability

All analyses presented in this report were conducted using publicly available data sources. VAERS data were obtained from publicly released datasets maintained by the Centers for Disease Control and Prevention (CDC) and the Food and Drug Administration (FDA). Publicly released CDC lot-distribution records were used to construct the denominator-adjusted analyses described in this report.

Analytical scripts, release documentation, supporting materials, and report outputs associated with this project are publicly available through the project repository:

<https://github.com/independentdatareview/vaers-covid-reporting-analysis>

The repository includes the source documents used to generate the findings presented in this report.

Methodology

Study Purpose and Interpretation

This analysis should be viewed as a hypothesis-generating signal-detection study rather than a causal safety study. The objective is to identify patterns warranting additional investigation, not to establish causation.

The findings are intended to support further review and replication, not to serve as definitive evidence regarding vaccine safety, product quality, or clinical risk.

Denominator Coverage

Denominator-adjusted calculations were limited to vaccine lots for which both adverse-event reporting data and publicly available lot-distribution records were available. Lots lacking sufficient denominator information could not be included in denominator-adjusted calculations.

As a result, the denominator-adjusted findings presented in this report should be interpreted as applying only to the matched lots analyzed. The analysis does not claim to represent all vaccine lots contained within the broader VAERS dataset.

Availability of denominator information remains a practical limitation for independent lot-level analysis. The release of lot-distribution data made the denominator-adjusted portion of this project possible.

Phase 1: Baseline VAERS Analysis

The first phase of the project analyzed COVID-related VAERS reports by manufacturer, lot number, state, age group, and symptom grouping.

Initial analyses examined:

- Total reports by manufacturer
- Serious reports by manufacturer
- Death reports by manufacturer
- Serious-reporting percentages
- Lot-level report counts
- Age-bucket patterns
- State-level reporting patterns
- Symptom clusters
- Targeted event categories: myocarditis, pericarditis, thrombosis, pulmonary embolism, deep vein thrombosis, cerebral venous sinus thrombosis, Guillain-Barré syndrome, syncope, and loss of consciousness

These analyses identified manufacturer-level differences and several lots that repeatedly appeared in elevated-reporting analyses.

Phase 2: Denominator Adjustment

The second phase incorporated publicly released CDC lot-distribution records. The purpose was to address the denominator problem in lot-level VAERS analysis.

The denominator-adjusted lot analysis was limited to matched Pfizer lots because the publicly released lot-distribution records incorporated into the denominator framework were Pfizer-specific. Pfizer lots with matched denominator information were therefore used to calculate reporting rates, expected serious-report counts, observed-versus-expected ratios, and confidence intervals.

For matched Pfizer lots, the following metrics were calculated:

- Total VAERS reports
- Serious reports
- Death reports
- Total doses shipped
- Reports per million doses shipped
- Serious reports per million doses shipped
- Deaths per million doses shipped
- Expected serious reports
- Observed-versus-expected serious-reporting ratio
- Approximate 95% confidence interval

Understanding Observed-Versus-Expected Ratios

The observed-versus-expected ratio (O/E) compares the number of serious reports observed for a vaccine lot with the number expected based on the overall baseline used in this analysis.

An O/E ratio of 1.0 indicates that the lot matched expectation. Values greater than 1.0 indicate more serious reports than expected, while values below 1.0 indicate fewer serious reports than expected.

For example, an O/E ratio of 2.0 indicates approximately twice as many serious reports as expected under the baseline. An O/E ratio of 5.0 indicates approximately five times as many serious reports as expected.

Approximate 95% confidence intervals were calculated to estimate statistical uncertainty. Confidence intervals that remain entirely above 1.0 suggest that an observed elevation is less likely to be explained by random variation alone within the framework of this analysis.

Findings

Manufacturer-Level Findings

The baseline analysis identified persistent reporting-pattern differences among manufacturers.

At the manufacturer level, earlier phases of the analysis found that Pfizer-associated reports were more enriched for myocarditis and pericarditis terms, while Janssen-associated reports were more enriched for thrombosis-related terms, cerebral venous sinus thrombosis, pulmonary embolism, deep vein thrombosis, Guillain-Barré syndrome, syncope, and loss of consciousness.

These manufacturer-level findings were broadly consistent with known safety-monitoring discussions in the medical and regulatory literature. The analysis was intended to identify reporting-pattern differences within VAERS rather than estimate clinical incidence rates.

Lot-Level Findings After Denominator Adjustment

Certain Pfizer lots remained elevated after accounting for doses shipped.

Lot	Doses Shipped	Serious Reports	Expected Serious Reports	O/E Ratio	Approx. 95% CI
GJ3277	3,930,955	579	114.68	5.049	4.638 to 5.460
EN6201	15,043,577	1,253	438.89	2.855	2.697 to 3.013
EN6200	13,890,331	1,102	405.25	2.719	2.559 to 2.880
EN6202	13,619,726	1,070	397.35	2.693	2.531 to 2.854
ER2613	15,392,071	926	449.06	2.062	1.929 to 2.195

The largest denominator-adjusted serious-reporting elevation observed in this analysis was lot GJ3277. It had 579 serious reports compared with 114.68 expected serious reports, producing an O/E ratio of 5.049. The confidence interval remained well above 1.0.

The EN6200, EN6201, and EN6202 lots also remained elevated after denominator adjustment. Despite distribution volumes ranging from approximately 13.6 million to 15.0 million doses shipped, their serious-reporting frequencies remained elevated relative to the Pfizer baseline used in this analysis.

Shipped-dose adjustment reduced several apparent signals but did not eliminate all observed reporting differences.

Signals That Diminished After Denominator Adjustment

Some lots that appeared notable when examined through raw report counts became less remarkable after denominator adjustment.

This outcome demonstrates why denominator information is necessary for lot-level analysis. Some apparent signals weakened substantially once distribution volume was incorporated into the analysis, indicating that raw report counts alone can overstate the significance of certain lots. The fact that some signals diminished while others remained elevated supports the use of denominator-adjusted analysis.

Relationship to Existing Safety Literature

The findings in this report should be interpreted within the broader COVID-19 vaccine safety-monitoring literature.

VAERS is a passive surveillance system and does not establish causation by itself. Passive surveillance systems are routinely used for signal detection, and several COVID-19 vaccine safety signals initially identified or monitored through post-market surveillance were later discussed in official safety communications and peer-reviewed publications.

The myocarditis and pericarditis findings in this project are consistent with broader literature identifying myocarditis as a rare but serious adverse event reported after mRNA COVID-19 vaccination, particularly among younger males.

The Janssen-associated thrombosis and Guillain-Barré findings are also consistent with prior CDC, FDA, and peer-reviewed safety discussions identifying thrombosis with thrombocytopenia syndrome and Guillain-Barré syndrome as rare but serious safety concerns following Janssen COVID-19 vaccination.

This report does not rely on those publications to prove the lot-level findings. Those publications instead provide context showing that multiple event categories identified in this analysis have also been recognized in official or peer-reviewed safety-monitoring literature.

The primary contribution of this report is narrower: it evaluates whether certain lot-level VAERS reporting rates remain elevated after incorporating lot-distribution denominator data.

Understanding the Findings in Plain English

One of the most common criticisms of VAERS-based analysis is that raw report counts can be misleading. A vaccine lot distributed across a large portion of the country would be expected to generate more reports than a lot distributed to a smaller population, even if the underlying reporting rate were identical.

Consider the following hypothetical example:

- Lot A generated 500 reports after 10 million doses were distributed.
- Lot B generated 100 reports after 500,000 doses were distributed.

Lot A would appear more concerning when examining raw counts alone. After adjusting for distribution volume, Lot B would have the higher reporting rate.

The denominator-adjusted analysis performed in this report applies that same logic to matched Pfizer vaccine lots for which publicly released CDC distribution data were available.

The results showed that some apparent reporting signals diminished after adjustment for shipped-dose volume, indicating that distribution size explained some differences observed in raw VAERS counts. Other reporting patterns remained elevated after adjustment.

These results do not establish causation, clinical incidence, manufacturing defects, or product-quality differences. Nor do they exclude those possibilities. They indicate that incorporation of shipped-dose denominators materially changed the interpretation of apparent reporting differences and narrowed the field of potential signals requiring further investigation.

What the Analysis Can Reasonably Tell Us

The analysis identified notable lot-level reporting differences within the matched Pfizer data examined in this project. Incorporating shipped-dose denominators materially changed the interpretation of several apparent reporting signals, reducing some differences while leaving others elevated.

The results suggest that denominator information provides a more informative framework for lot-level analysis than raw report counts alone. Within the analyzed data, adjustment for shipped-dose volume did not eliminate all observed reporting differences.

The observed differences do not constitute evidence of biological causation, manufacturing defects, product contamination, formulation differences, clinical incidence rates, or individual injury determinations. Rather, they indicate that certain differences remained elevated after denominator adjustment and remain appropriate candidates for further evaluation using stronger independent datasets.

Underreporting Considerations

VAERS is a passive reporting system. Passive systems are widely understood to capture only a portion of adverse events, with reporting completeness varying by event type, severity, public awareness, provider behavior, and other factors.

This limitation affects interpretation of absolute event counts. If events are not fully captured, the number reflected in VAERS may be lower than the number that actually occurred.

Incomplete reporting alone, however, does not automatically explain why certain lots exhibit substantially higher reporting frequencies than other lots. If reporting gaps were uniform across lots, they would affect absolute counts but would not necessarily erase relative differences. If reporting varied by geography, population, time period, provider behavior, or public awareness, it could influence lot-level comparisons.

For that reason, incomplete reporting remains an important limitation, but it does not remove the need to investigate denominator-adjusted lot-level differences that persist within the available data.

Regulatory and Safety Context

This report does not evaluate FDA authorization decisions, clinical-trial outcomes, manufacturing quality controls, or regulatory compliance. The analysis is limited to publicly available adverse-event reports and publicly released lot-distribution records.

Within that limited scope, the findings are relevant to post-market safety monitoring. Passive surveillance systems such as VAERS are designed to identify patterns that may warrant additional review rather than establish causation directly.

The results presented here suggest that incorporating denominator information can materially affect the interpretation of lot-level reporting differences. As a result, access to denominator data remains an important component of reliable independent evaluation and replication.

Limitations

Like all analyses based on publicly available surveillance data, this study is subject to important methodological and data-related limitations. These limitations should be considered when interpreting the findings and evaluating the conclusions presented in this report.

VAERS Does Not Establish Causation

VAERS is a passive surveillance system. A report submitted to VAERS does not prove that a vaccine caused the reported event.

Doses Shipped Are Not Doses Administered

The denominator analysis used doses shipped rather than confirmed doses administered. Doses shipped provide a useful distribution denominator, but they are not equivalent to administered-dose records.

Reporting Frequency Is Not Incidence

The rates calculated in this report are reporting rates, not clinical incidence rates. They should not be interpreted as true population risk.

Lot Data Required Cleaning and Normalization

Lot identifiers in public adverse-event datasets are often inconsistent. Lot-number fields may contain typographical errors, formatting issues, missing values, or malformed entries. Normalization was necessary to match records across datasets.

Duplicate and Follow-Up Reporting Considerations

VAERS is a passive surveillance system and may contain duplicate submissions, follow-up reports, amended records, or reports describing the same clinical event from multiple sources.

The analysis relied on publicly available VAERS records as released and did not independently validate individual case reports. Any residual duplication or reporting inconsistencies present within the underlying data could influence observed reporting frequencies.

This limitation applies broadly to many analyses that rely on publicly available passive-surveillance datasets.

Multiple Comparisons

This analysis did not apply a formal multiple-comparison correction or false-discovery-rate adjustment. Consequently, elevated lots identified in this report should be viewed as candidates for further investigation and replication rather than definitive findings.

Because many lots were evaluated simultaneously, some elevated observations may represent chance findings rather than underlying differences. Future work should evaluate whether the observed signals persist under alternative statistical approaches and across independent datasets.

Residual Confounding May Remain

Lot-level reporting patterns may be influenced by factors not fully accounted for in this analysis, including geography, age distribution, reporting behavior, time period, public awareness, provider reporting practices, and distribution channel.

The Analysis Is Limited to Available Public Data

The analysis did not include internal manufacturer records, lot-testing records, complete administered-dose data, active-surveillance datasets, or clinical case review records.

Future Research Priorities

This analysis should be viewed as an initial investigation rather than a definitive assessment. Confirmed administered-dose denominators by lot remain one of the most important unresolved data elements.

Future work could be strengthened through access to:

- Confirmed doses administered by lot
- State immunization registry data
- V-Safe data
- Vaccine Safety Datalink data
- FDA BEST or Sentinel data
- CMS and Medicare claims data
- Private insurance claims data
- Manufacturing and quality-control records
- Lot-specific testing records
- Clinical follow-up investigation records

Access to administered-dose records would allow future analyses to evaluate whether observed denominator-adjusted patterns persist when distribution volume is replaced with actual administration volume.

Replication across independent datasets remains one of the strongest ways to determine whether observed signals reflect genuine safety patterns, reporting artifacts, or other explanations.

Conclusion

This report examined whether apparent COVID-19 vaccine lot-level differences remained visible after adjustment for shipped-dose volume using publicly available VAERS and CDC lot-distribution data.

The analysis found that denominator adjustment materially changed the interpretation of several apparent reporting signals. Some lot-level differences diminished after adjustment, indicating that distribution volume explained part of the variation observed in raw report counts.

Within the matched Pfizer lots analyzed, shipped-dose adjustment did not eliminate all observed reporting differences.

The results should not be interpreted as evidence of causation, clinical incidence, manufacturing defects, product-quality differences, or individual injury determinations. VAERS remains a passive surveillance system subject to important limitations, and shipped-dose counts remain an imperfect denominator.

The primary contribution of this analysis is methodological. Incorporating denominator information provides a more informative framework for evaluating **lot-level reporting rates** than raw report counts alone. The signals identified in this analysis warrant replication and further evaluation using administered-dose denominators, active-surveillance systems, claims databases, and other independent data sources.

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U.S. Food and Drug Administration. FDA Approves Required Updated Warning in Labeling of mRNA COVID-19 Vaccines Regarding Myocarditis and Pericarditis Following Vaccination. June 25, 2025.

U.S. Food and Drug Administration. Janssen COVID-19 Vaccine Emergency Use Authorization Materials and Safety Communications.

Publicly released CDC lot-distribution records incorporated into the VAERS COVID Reporting Pattern Analysis repository.

Independent Researcher, VAERS COVID Reporting Pattern Analysis, Version 2.0 Public Research Release. GitHub Repository and Release Package. June 2026.

Appendix A – Lot Number Normalization and Matching Methodology

Public VAERS lot identifiers contain substantial variability and data-quality issues. To improve matching between VAERS reports and publicly released lot-distribution records, the following normalization procedures were applied:

1. Conversion of all lot identifiers to uppercase.
2. Removal of leading and trailing whitespace.
3. Removal of non-printable characters.
4. Standardization of common formatting inconsistencies where appropriate.
5. Exclusion of blank, missing, or obviously invalid lot entries.
6. Preservation of original identifiers where ambiguity existed.
7. Matching against denominator datasets using normalized lot identifiers.

No attempt was made to infer unknown lot numbers or reconstruct incomplete identifiers beyond basic normalization procedures.

All matching rules, scripts, and outputs are included within the public repository to support independent review and replication.

Appendix B – Reproducibility Information

Software Environment

The analyses described in this report were performed using publicly available software tools, including:

- Python 3.13.13
- DuckDB 1.5.3
- Pandas 3.0.3
- Standard Python statistical and data-processing libraries

Data Sources

The analysis relied exclusively on publicly available datasets and records, including:

- VAERS report data
- VAERS vaccine data
- VAERS symptom data
- Publicly released CDC lot-distribution records

No proprietary manufacturer datasets, internal FDA records, active-surveillance databases, clinical case files, or non-public government records were used.

Repository and Release Materials

Analytical scripts, release documentation, supporting materials, and report outputs are available through the project repository:

<https://github.com/independentdatareview/vaers-covid-reporting-analysis>

Repository contents include:

- Source documentation
- Analytical scripts
- Release notes
- Research reports
- Supporting appendices
- Output files used in the preparation of this report

Release Version

Report Version: 2.0 Research Release; Publication Date: June 2026

Replication

Readers may review the repository materials, methodology, assumptions, normalization procedures, calculations, and limitations independently. Replication using alternative datasets, administered-dose denominators, active-surveillance systems, claims databases, and other independent sources of evidence is encouraged.